



Rounak Das

Roll No.: 20243585

B.Tech, Computer Science & Engineering

2nd Year | Batch 2024–2028

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EDUCATION

- **Motilal Nehru National Institute of Technology (MNNIT) Allahabad** Expected May 2028
Bachelor of Technology in Computer Science & Engineering CPI: 7.12
- **Indian School Muscat, Muscat, Oman** Graduated Apr 2024
Higher Secondary Examination – CBSE Percentage: 95.4%

EXPERIENCE

- **Freelance Game Developer** May 2021 – Present
Lead Programmer Roblox Platform
 - Led end-to-end development of 5+ shipped titles, including games that reached 50M+ total visits and sustained 1,500+ concurrent players.
 - Engineered fault-tolerant datastore pipelines with rollback safeguards, maintaining data integrity and zero-loss persistence under peak concurrent load.
 - Implemented reusable, modular combat and physics frameworks covering airborne, aquatic, and ground vehicle mechanics.
 - Designed NPC AI systems using A* pathfinding and finite state machines that scaled adaptively across varied map layouts and game genres.

PERSONAL PROJECTS

- **Game Boy Color Emulator** 2023
C, Systems Architecture
 - Implemented a CPU emulator in C, correctly handling the full instruction set, interrupt dispatch logic, and memory-mapped I/O.
 - Passed the Blargg CPU instruction test suite in full, validating correctness against a standardized benchmark covering edge cases in real Game Boy Color ROM execution.
- **Stock Price Prediction via Time Series Analysis** 2025–2026
Python, ARIMA, SARIMA, Pandas, NumPy, Streamlit, Plotly
 - Built a modular Streamlit forecasting app implementing 5 models (ARIMA, SARIMA, Seasonal Naive, Moving Average, Historic Mean) on AAPL, META, and MSFT stock data, achieving 98.4% directional accuracy on a held-out test set.
 - Automated ARIMA/SARIMA hyperparameter selection (p, d, q) from ACF/PACF analysis and ADF stationarity tests, eliminating hardcoded parameters and enabling data-driven model configuration.
- **Open Source Contributions** Oct 2025 – Ongoing
C, R, Data Visualization, Embedded Systems
 - Implemented broadcasting for tensor `matmul` on 3D/4D matrices in cTensor (C tensor library); extended operator to handle arbitrary batch dimensions with a comprehensive test suite; PR merged with 819 additions.
 - Traced a root-cause compiler bug in animint2 (R interactive data visualization library): `compute_domains()` fails for `geom_bar` due to a column mismatch in `built_data`; added a regression test and diagnostic warning surfacing the missing columns.

TECHNICAL SKILLS AND INTERESTS

Languages: Python, C/C++, Lua, JavaScript/TypeScript, Java, SQL

Developer Tools: Git/GitHub, Neovim, Roblox Studio, Docker, Linux Terminal, CI/CD (GitHub Actions)

Frameworks & Libraries: NumPy, Pandas, Scikit-learn, Statsmodels (ARIMA), Matplotlib, Streamlit, Plotly, SDL

Areas of Interest: Systems Programming, Compiler Design, Data Science, Game Development, Embedded Systems

Coursework: Data Structures & Algorithms, Object-Oriented Programming, System Design

ACHIEVEMENTS

- **Chairman's Award**, For achieving the highest grade (A1) in all subjects. 2024
- **Principal's Award**, Outstanding overall scholastic performance in Class 12. 2024